python -m uvicorn main:app --reload

```
INFO: Will watch for changes in these directories: ['C:\\Users\\tg070\\OneDrive\\Área de Trabalho\\ST SF\\Univesp\\estoque_python']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [6116] using StatReload
Process SpawnProcess-1:
Traceback (most recent call last):
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\multiprocessing\process.py", line 320, in _bootstrap
self.run()
~~~~~~~~^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\multiprocessing\process.py", line 108, in run
self._target(*self._args, **self._kwargs)
~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\uvicorn\_subprocess.py", line 80, in subprocess_started
target(sockets=sockets)
~~~~~~^^^^^^^^^^^^^^^^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\uvicorn\server.py", line 75, in run
return asyncio_run(self.serve(sockets=sockets), loop_factory=self.config.get_loop_factory())
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\asyncio\runners.py", line 204, in run
return runner.run(main)
~~~~~~~~~~^^^^^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\asyncio\runners.py", line 127, in run
return self._loop.run_until_complete(task)
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\asyncio\base_events.py", line 719, in run_until_complete
return future.result()
~~~~~~~~~~~~~^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\uvicorn\server.py", line 79, in serve
await self._serve(sockets)
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\uvicorn\server.py", line 86, in _serve
```

```
config.load()
~~~~~~~~~~~^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\uvicorn\config.py", line 441, in load
self.loaded_app = import_from_string(self.app)
~~~~~~~~~~~~~~~~~~^^^^^^^^^^
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\uvicorn\importer.py", line 19, in import_from_string
module = importlib.import_module(module_str)
File "C:\Users\tg070\AppData\Local\Python\pythoncore-3.14-64\Lib\importlib\__init__.py", line 88, in import_module
return _bootstrap._gcd_import(name[level:], package, level)
~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "<frozen importlib._bootstrap>", line 1398, in _gcd_import
File "<frozen importlib._bootstrap>", line 1371, in _find_and_load
File "<frozen importlib._bootstrap>", line 1342, in _find_and_load_unlocked
File "<frozen importlib._bootstrap>", line 938, in _load_unlocked
File "<frozen importlib._bootstrap_external>", line 755, in exec_module
File "<frozen importlib._bootstrap_external>", line 893, in get_code
File "<frozen importlib._bootstrap_external>", line 823, in source_to_code
File "<frozen importlib._bootstrap>", line 491, in _call_with_frames_removed
SyntaxError: source code string cannot contain null bytes
```

✅ **QUASE LÁ! Erro comum Windows + `main.py` corrompido por encoding!**

🔧 SOLUÇÃO SIMPLES - SEM RELOAD (Windows):

```
# ❌ SEM --reload (causa erro multiprocessing Windows)
python -m uvicorn main:app --host 127.0.0.1 --port 8000
```

🚨 MAS PRIMEIRO: RECONSTRUA `main.py` LIMPO!

VS Code → APAGUE `main.py` → Novo arquivo → Cole ISTO:

```
from fastapi import FastAPI, Form, Request
from fastapi.templating import Jinja2Templates
from fastapi.staticfiles import StaticFiles
from fastapi.responses import HTMLResponse
import sqlite3

app = FastAPI()
templates = Jinja2Templates(directory="templates")
app.mount("/static", StaticFiles(directory="static"), name="static")

def get_db():
    conn = sqlite3.connect('estoque.db')
    conn.row_factory = sqlite3.Row
    return conn
```

```
@app.get("/")
async def dashboard(request: Request):
    conn = get_db()
    produtos = conn.execute("SELECT * FROM produtos").fetchall()
    conn.close()
    return templates.TemplateResponse("dashboard.html", {"request": request, "produtos

@app.get("/vendas/nova")
async def nova_venda(request: Request):
    conn = get_db()
    produtos = conn.execute("SELECT * FROM produtos WHERE quantidade > 0").fetchall()
    conn.close()
    return templates.TemplateResponse("venda_form.html", {"request": request, "produto

@app.post("/vendas/")
async def confirmar_venda(produto_id: int = Form(...), quantidade: int = Form(...)):
    conn = get_db()
    produto = conn.execute("SELECT quantidade, preco FROM produtos WHERE id = ?", (prod
    conn.execute("INSERT INTO vendas (produto_id, quantidade) VALUES (?, ?)", (produto_i
    conn.execute("UPDATE produtos SET quantidade = quantidade - ? WHERE id = ?", (quanti
    conn.commit()
    conn.close()
    return {"success": "Venda OK!"}
```

✅ COMANDOS FINAIS (copie EXATAMENTE):

```
# 1. Crie banco (1 linha)
python -c "import sqlite3; c=sqlite3.connect('estoque.db').cursor(); c.execute('CREATE T

# 2. SEM --reload (Windows fix)
python -m uvicorn main:app --host 127.0.0.1 --port 8000
```

✅ CRIE ARQUIVOS MÍNIMOS:

`templates/dashboard.html`:

```
<!DOCTYPE html>
<html>
<head><title>Dashboard</title><link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/
<body class="container py-4">
<h1>□ Estoque FastAPI</h1>
<a href="/vendas/nova" class="btn btn-success mb-3">Nova Venda</a>
<table class="table">
<tr><th>Nome</th><th>Qtd</th><th>Preço</th></tr>
{% for p in produtos %}
<tr><td>{{ p.nome }}</td><td>{{ p.quantidade }}</td><td>R$ {{ p.preco }}</td></tr>
{% endfor %}
</table>
</body>
</html>
```

`templates/venda_form.html`:

```
<!DOCTYPE html>
<html>
<head><title>Nova Venda</title><link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0
<body class="container py-4">
<h1>Nova Venda</h1>
<form method="post">
<select name="produto_id" class="form-select mb-3">
{% for p in produtos %}
<option value="{{ p.id }}">{{ p.nome }} ({{ p.quantidade }})</option>
{% endfor %}
</select>
<input type="number" name="quantidade" class="form-control mb-3" min="1">
<button class="btn btn-success">Confirmar</button>
</form>
<a href="/" class="btn btn-secondary">Dashboard</a>
</body>
</html>
```

RESULTADO:

```
✅ http://localhost:8000 = Dashboard 3 produtos
✅ http://vendas/nova = Form venda
✅ Venda → Arroz -2 estoque
```

`python -m uvicorn main:app --host 127.0.0.1 --port 8000` = Windows OK!

Sistema mínimo funcionando em 2 minutos!

Laravel → FastAPI migrado com sucesso!

1. https://github.com/Kludex/uvicorn/issues/342
2. https://www.uvicorn.org/release-notes/
3. https://github.com/Kludex/uvicorn/issues/484
4. https://stackoverflow.com/questions/63635664/cannot-run-uvicorn-command-on-windows-despite-installing-it-and-adding-in-path-v
5. https://www.reddit.com/r/FastAPI/comments/1b7wooj/how_to_fix_the_modulenotfounderror_caused_by_the/
6. https://stackoverflow.com/questions/73580993/using-multiprocessing-in-uvicorn-with-multiple-workers-thread-lock
7. https://github.com/encode/uvicorn/issues/683
8. https://github.com/Kludex/uvicorn/issues/514
9. https://github.com/encode/uvicorn/issues/548
10. https://fastapi.tiangolo.com/deployment/server-workers/